

$$A = [a_1 \dots a_n] \quad a_i \in \mathbb{R}^n$$

QR complete

$$A = Q \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix} z^T$$

$$Q^T A z = \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

$m \times m \quad m \times n \quad n \times n$

$$\begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

$$r = \text{rank}(A)$$

Triangular Superior
(Householder + Permut)
Bidiagonal
(Householder + Permutaci)
Alternado

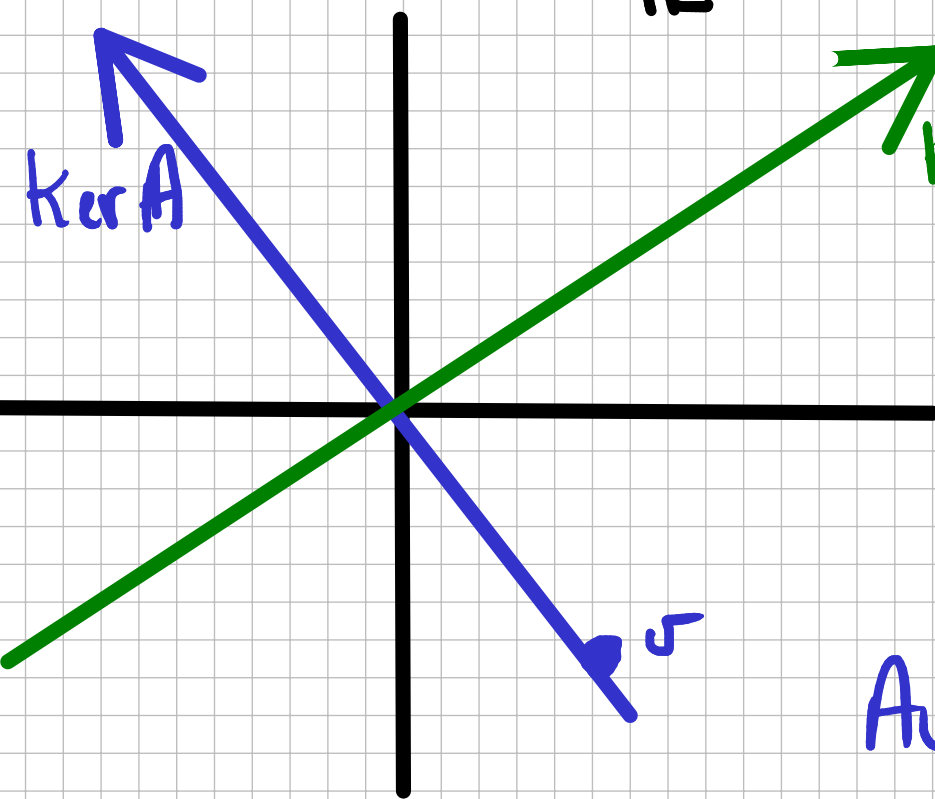
$$\begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_{11} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 & z_2 \end{bmatrix}^T = Q_1 R_{11} z_1$$

factorization QR

Permutaci col.

[...]

[...]



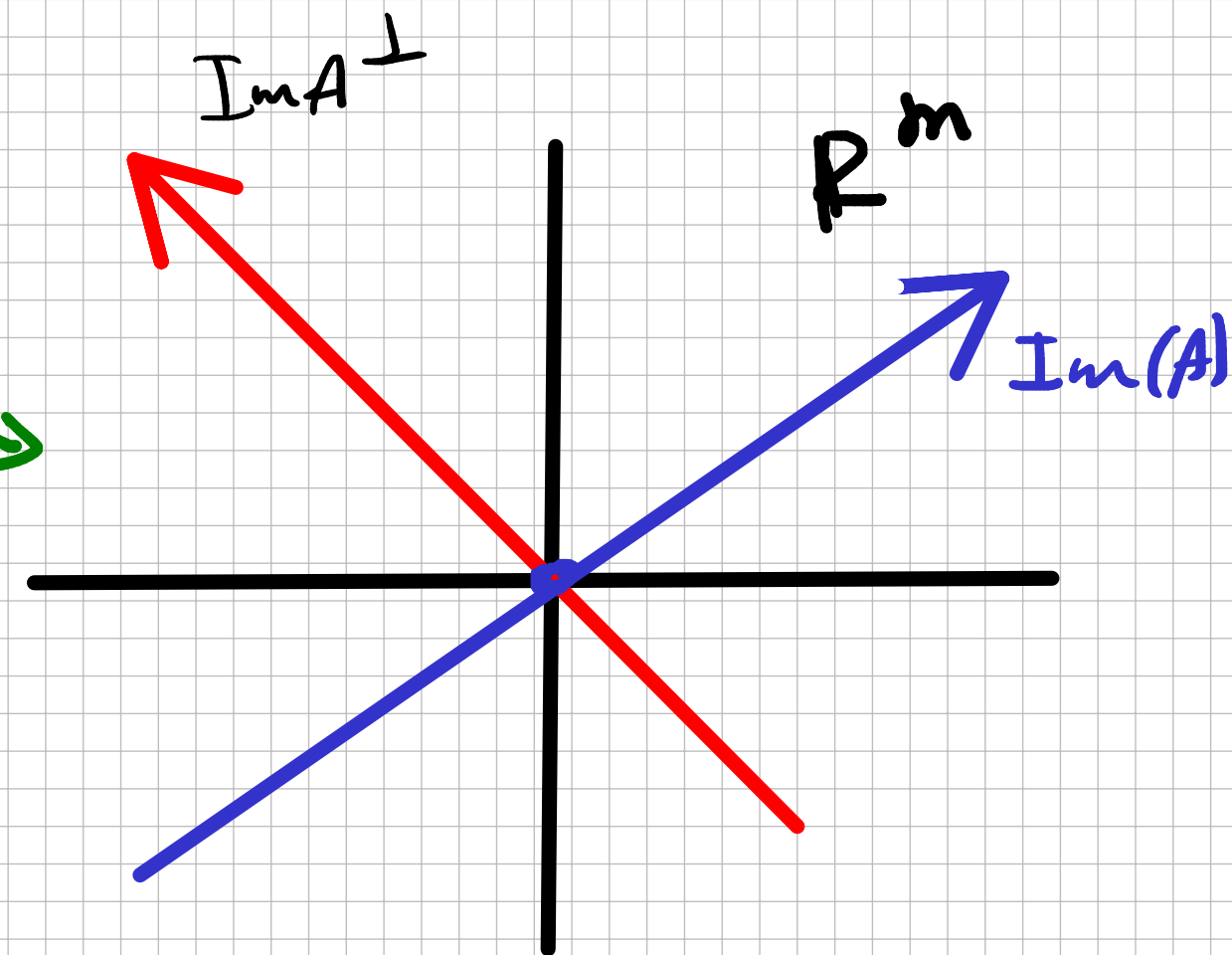
$$A u = 0$$

$$\text{Im}(A) = \text{span } Q_1$$

$$\text{Im}(A)^\perp = \text{span } Q_2$$

$$\ker(A) = \text{span } z_2$$

$$\ker(A)^\perp = \text{span } z_1$$



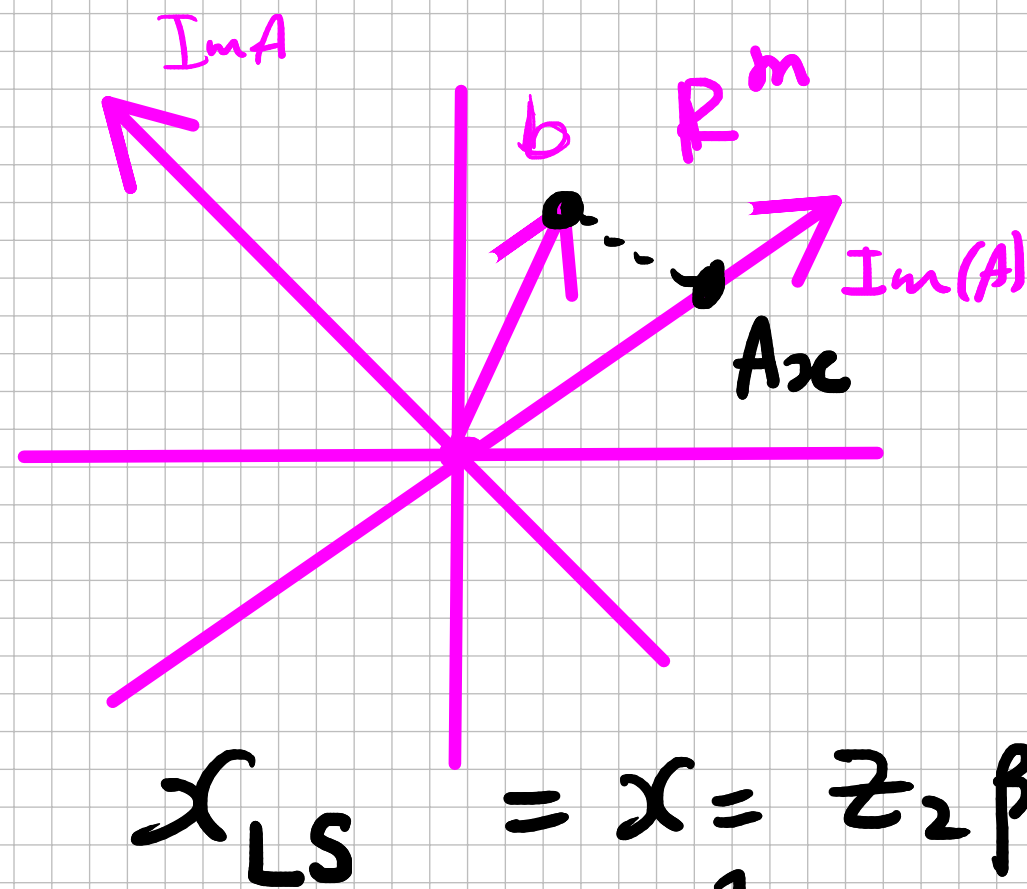
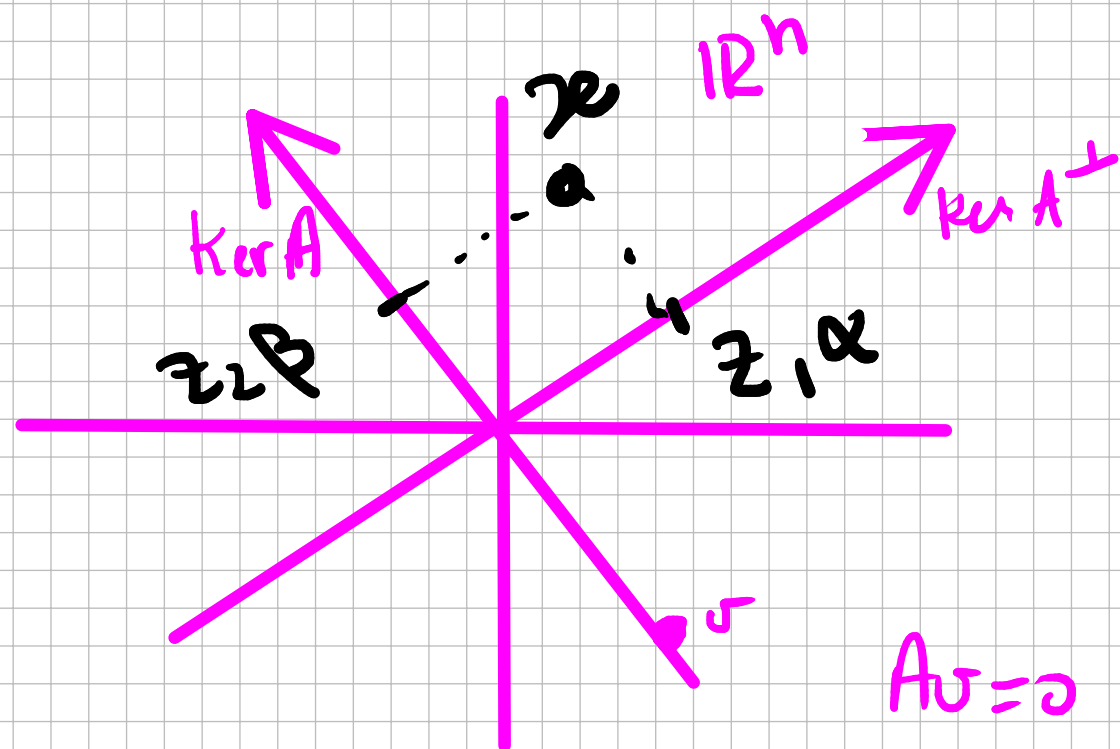
Option 2

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2 = \min_{\alpha \in \mathbb{R}^r} \sqrt{\|Q_2^T b\|^2 + \|\underline{R_{11}} \alpha - Q_1^T b\|_2^2}$$

$$\|Ax - b\|_2^2 = \|Q_2^T b\|^2 + \|\underline{R_{11}} \alpha - Q_1^T b\|_2^2$$

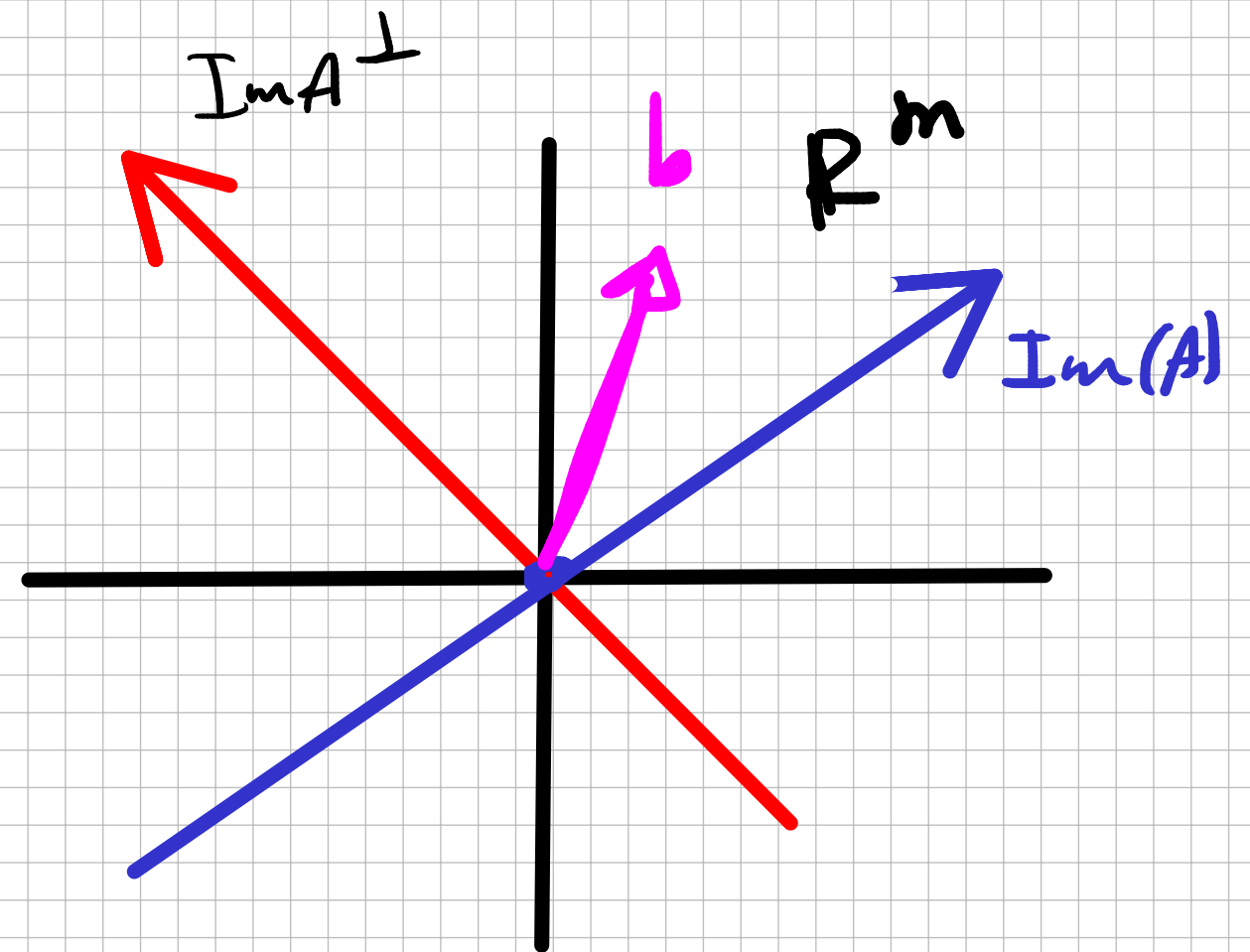
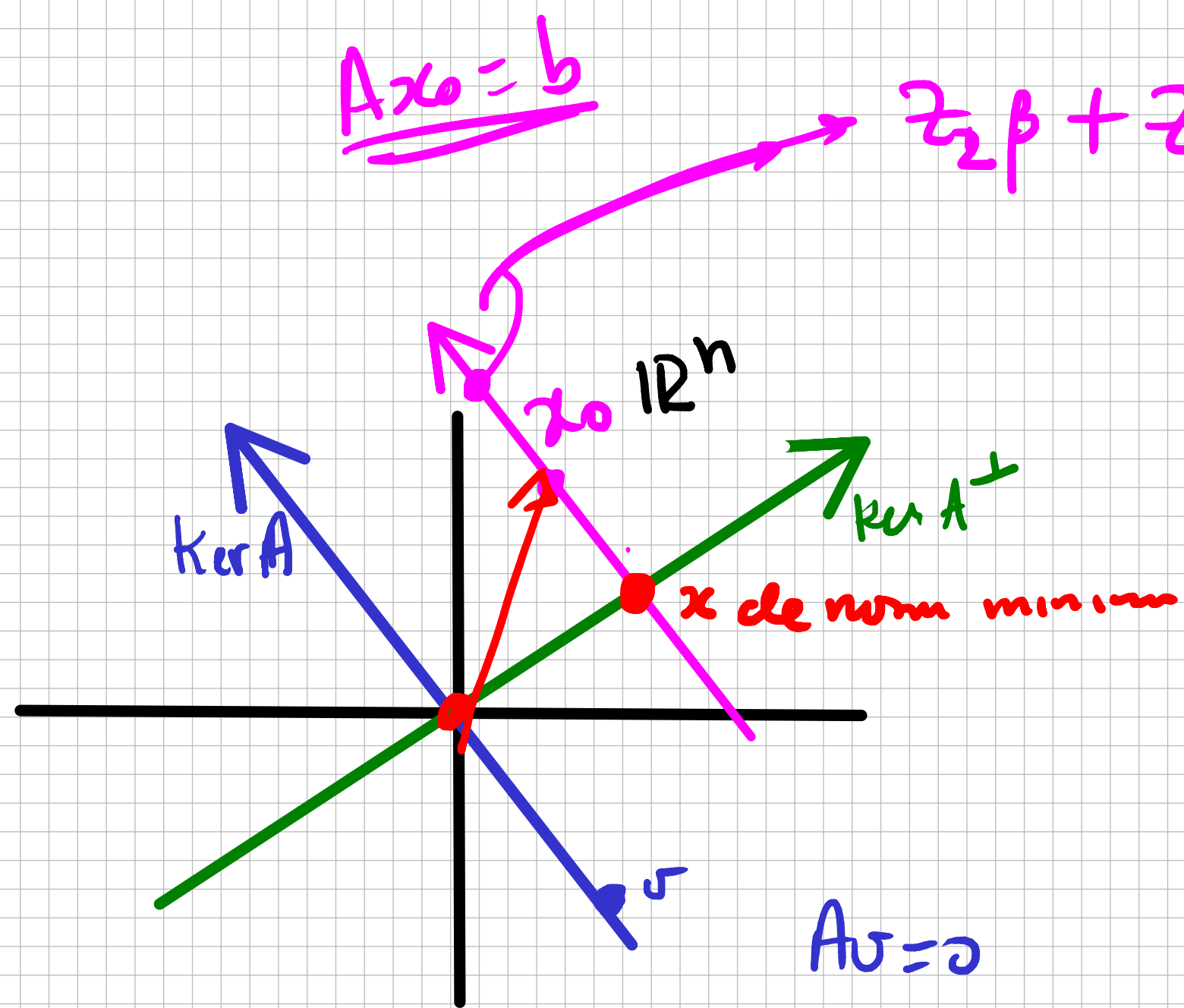
$$\downarrow$$

$$\alpha = \underline{R_{11}^{-1}} Q_1^T b$$



$$x_{LS} = x = z_2 \beta + z_1 \alpha$$

$$\|x\|^2 = \|z_2 \beta\|_2^2 + \|z_1 \alpha\|_2^2$$



Problema 3

$$1 + \alpha x + \beta x^2$$

$\bar{x}$ tal que

$r = b - Ax$ sea

ortogonal a filas de A^T

$$A^T r = 0 \quad A^T (b - A\bar{x}) = 0$$

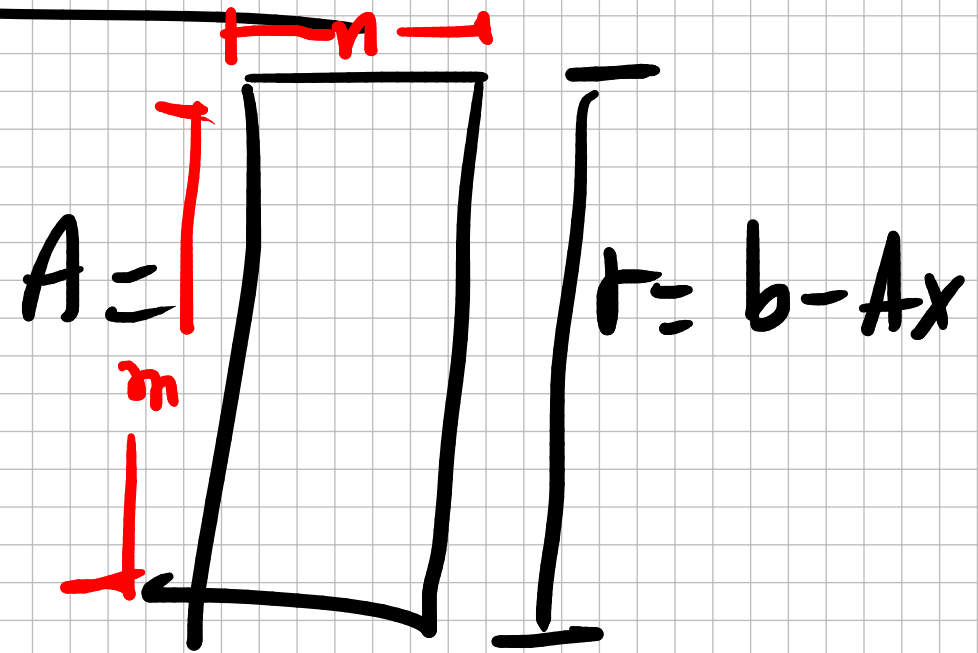
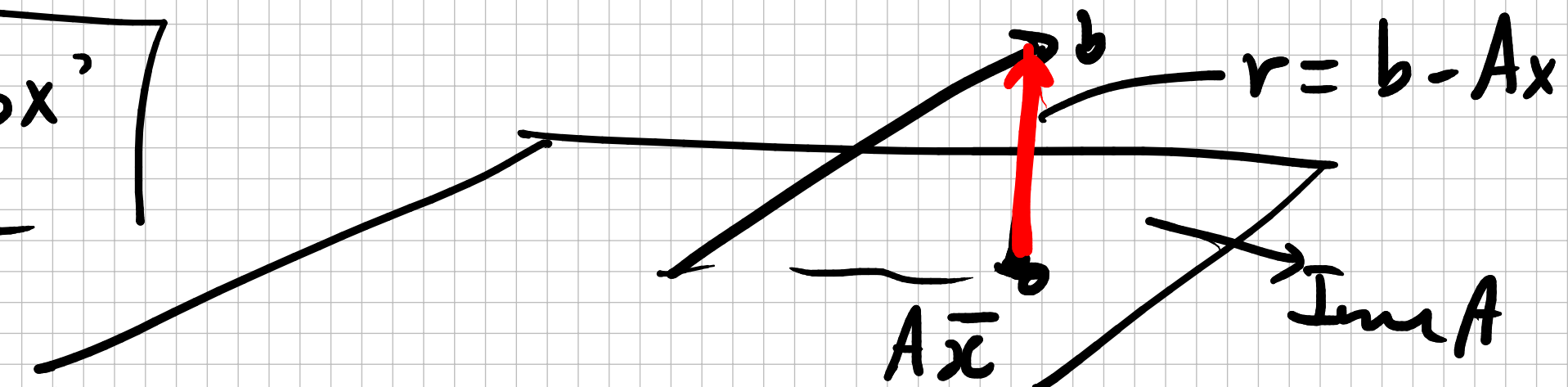
Ecuaciones
Normales

$$A^T A \bar{x} = A^T b$$

No singular si $\text{rang } A = n$.

⊥

$$\text{si } A v = 0, \quad A^T A v = 0$$



$$x = (A^T A)^{-1} A^T b$$

$$= x^* = x_{LS}$$

Opçon 4

Singular Value Decomposition. SVD

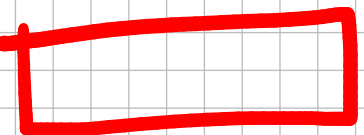
$$\begin{matrix} m \times m & m \times n & n \times n \end{matrix} \quad U^T A V = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p)_{m \times n}$$

$$\begin{cases} U^T U = I \\ V^T V = I \end{cases}$$

$A \rightarrow$  $\begin{matrix} p=n \\ p=m \end{matrix}$

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > \sigma_{r+1} = 0 = \dots = \sigma_p = 0$$

$$r = \text{rango}(A) = \text{rank}(A)$$

$A =$  $p = \min\{m, n\}$

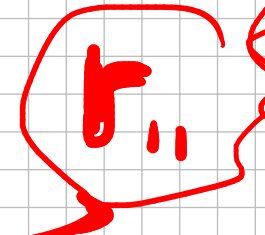
$$m \geq n$$

$$\Sigma = \begin{pmatrix} \text{diag}(\sigma_1, \dots, \sigma_r) \\ 0 \end{pmatrix}$$

$$m \leq n$$

$$\Sigma = \begin{pmatrix} \text{diag}(\sigma_1, \dots, \sigma_r) & 0 \end{pmatrix}$$

$$\max_x \frac{\|Ax\|}{\|x\|} \geq \frac{\|Ae_1\|}{\|e_1\|}$$

σ_1 

Existe S.V.D.

$$\left\| A \left(\frac{\hat{x}}{\|\hat{x}\|_2} \right) \right\|_2$$

$$\underline{\sigma_1} = \|A\|_2 = \text{opnorm}(A) = \max_{\hat{x} \in \mathbb{R}^n, \|\hat{x}\|_2=1} \frac{\|A\hat{x}\|_2}{\|\hat{x}\|_2} = \max_{\|u\|_2=1} \|Au\|_2$$

$$\left[\begin{array}{l} u \mapsto \|Au\|_2 \\ \text{continuum} \\ u \in \text{compacto} \end{array} \right]$$

existe x , $\|x\|=1$,

$$\|Ax\|_2 = \sigma_1$$

$$y = \frac{Ax}{\sigma_1}, \quad \|y\| = \frac{\|Ax\|}{\sigma_1} = 1.$$

$$U = [y, U_2]$$

$$V = [x, V_2]$$

$$A_1 = U^T A V$$

$$= \begin{bmatrix} y^T \\ U_2^T \end{bmatrix} A \begin{bmatrix} x & V_2 \end{bmatrix}$$

$$= \begin{bmatrix} \overbrace{y^T A x}^{\sigma_1} & \overbrace{y^T A V_2}^{\omega} \\ \underbrace{U_2^T A x}_{\sigma_2 y} & \underbrace{U_2^T A V_2}_{\omega} \end{bmatrix}$$

$$A_1 = U^T A V = \begin{bmatrix} \sigma_1 & \omega \\ 0 & U_2^T A V_2 \end{bmatrix}$$

$m \times n$ $m \times n$ $n \times n$ $m-1 \times n-1$

ω



$$A_1 = U^T A V = \begin{bmatrix} \sigma_1 & 0 \\ 0 & B \end{bmatrix}$$

$\omega = 0$

$$A_1 \begin{pmatrix} \sigma_1 \\ \omega^T \end{pmatrix} = \begin{bmatrix} \sigma_1 & \omega \\ 0 & U_2^T A V_2 \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \omega^T \end{bmatrix} = \begin{bmatrix} \sigma_1^2 + \|\omega\|^2 \\ U_2^T A V_2 \omega^T \end{bmatrix}$$

$$\Rightarrow \left\| A_1 \begin{pmatrix} \sigma_1 \\ \omega^T \end{pmatrix} \right\|_2^2 \geq (\sigma_1^2 + \|\omega\|^2)^2$$

$$\frac{\|A_1 \bar{z}\|_2^2}{\|\bar{z}\|_2^2} \geq \sigma_1^2 + \|\omega\|^2$$

$\rightarrow \|\bar{z}\| = \sigma_1^2 + \|\omega\|^2$

$\bar{z} = V^T x$

$x = V \bar{z}$

$\|A_1 \bar{z}\|_2$

$\frac{\|A V \bar{z}\|_2}{\|\bar{z}\|_2}$

$\|A_1 \bar{z}\|^2 = \bar{z}^T A_1^T A_1 \bar{z} = \bar{z}^T V^T A^T A V \bar{z} = \|A(V \bar{z})\|_2^2$

$\sigma_1 = \max_{\substack{x \in \mathbb{R}^n \\ x \neq 0}} \frac{\|Ax\|_2}{\|x\|_2}$

$= \max_{\substack{z \in \mathbb{R}^n \\ z \neq 0}} \frac{\|A V z\|_2}{\|z\|_2}$

$$\sigma_1 = \max \frac{\|A \bar{z}\|}{\|\bar{z}\|} \geq \frac{\|A_1 \bar{z}\|}{\|\bar{z}\|} \geq \sqrt{\sigma_1^2 + \|\omega\|^2} \Rightarrow \|\omega\|^2 = 0$$

$$A_1 = U^T A V = \begin{bmatrix} \sigma & 0 \\ 0 & B \end{bmatrix}$$

Repetimos y encontramos $\hat{U}_1$ $\hat{V}_1$ tal q

$$\begin{pmatrix} 1 & 0 \\ 0 & \hat{U}_1 \end{pmatrix} U^T A V \begin{pmatrix} 1 & 0 \\ 0 & \hat{V}_1 \end{pmatrix} = \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & \boxed{C} \end{pmatrix}$$

continuamos de esta forma y obtenemos SUP.

$$U^T A V = \Sigma$$

$$A = U \Sigma V^T$$

$$\Sigma = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}$$

$$\rightarrow \Sigma = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}$$

$$U = \begin{bmatrix} | & & | \\ u_1 & u_2 & \dots & u_m \\ | & & | \end{bmatrix}$$

$$V = \begin{bmatrix} | & & | \\ v_1 & v_2 & \dots & v_n \\ | & & | \end{bmatrix}$$

$$A = U \Sigma V^T =$$

$$\sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T + 0$$

$$u_1 = \begin{bmatrix} u_{11} \\ u_{21} \\ u_{31} \\ \vdots \\ u_{m1} \end{bmatrix}$$

$$v_1^T = \begin{bmatrix} v_{11} & v_{21} & v_{31} & \dots & v_{n1} \end{bmatrix}$$

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$$

$$\sigma_{r+1} = 0 \dots = \sigma_p = 0$$

$$\sigma_1 > \sigma_2 > \sigma_3$$

$$A^{m \times n} = U \Sigma V^T = \underbrace{\sigma_1 u_1 v_1^T}_{\text{red underline}} + \underbrace{\sigma_2 u_2 v_2^T}_{\text{red underline}} + \dots + \underbrace{\sigma_r u_r v_r^T}_{\text{red underline}} + 0$$

$$Ax = \underbrace{\sigma_1}_{\text{red underline}} \underbrace{u_1 (v_1^T x)}_{\text{red underline}} + \sigma_2 u_2 (v_2^T x) + \dots$$

$$\boxed{A \vec{v}_j = \sigma_j u_j}$$

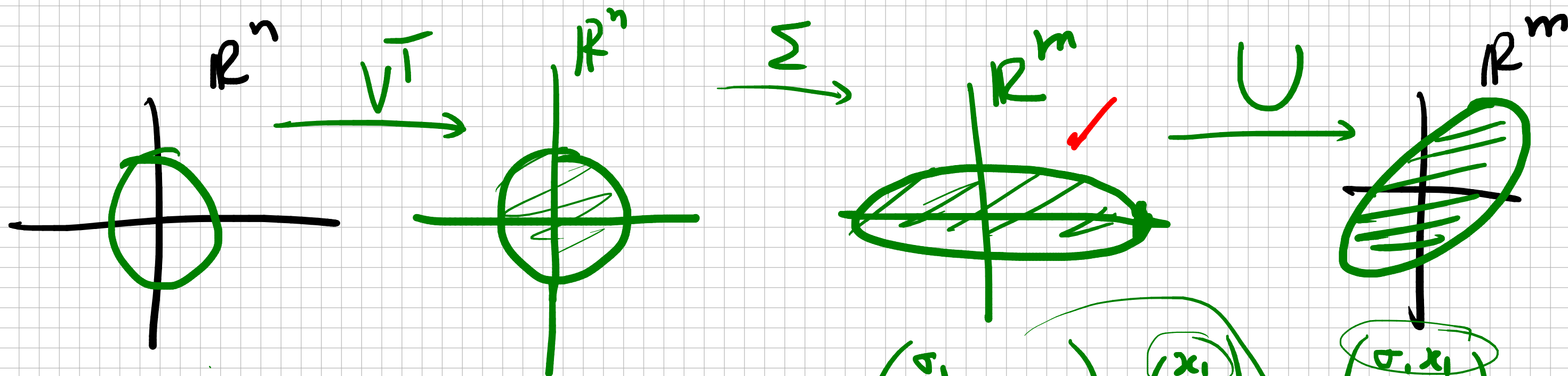
↑

$$u_k^T A = \sigma_k v_k^T$$

$$\underbrace{Ax = \lambda x}_{\text{black underline}}$$

$$A = U \Sigma V^T$$

$$A: \mathbb{R}^n \rightarrow \mathbb{R}^m$$



$$\Sigma x = \begin{pmatrix} 0 & \ddots & \sigma_m & 0 \\ & & & \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$$

$$\Sigma x = \begin{pmatrix} \sigma_1 & \sigma_2 & \dots & \sigma_n \\ & & & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \sigma_1 x_1 \\ \sigma_2 x_2 \\ \vdots \\ \sigma_n x_n \\ 0 \\ \vdots \end{pmatrix}$$

$$Ax = b$$

$$U \Sigma V^T x = b$$

$$\rightarrow \Sigma V^T x = U^T b$$

$$U_1 \Sigma_1 V_1^T x = b$$

$$\Sigma_1 V_1^T x = U_1^T b$$

$$\alpha = V_1^T x = \Sigma_1^{-1} U_1^T b$$

$$x = \underbrace{V_1}_{\text{green}} \alpha + V_2 \overset{0}{\beta} = [V_1 \ V_2] \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$V_1^T x = \underbrace{V_1^T V_1}_{\text{green}} \alpha + \underbrace{V_1^T V_2}_{\text{black}} \beta = \alpha$$

$$A = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \dots & \sigma_r & 0 & \dots & 0 \\ & & & & & \\ & & & 0 & & \\ & & & & & \\ & & & & & 0 \end{bmatrix} \begin{bmatrix} V_1 & V_2 \end{bmatrix}^T \quad \Sigma$$

$$= \begin{matrix} U_1 & \Sigma_1 & V_1^T \\ m \times r & r \times r & r \times n \end{matrix}$$

$$V_1^T V_1 = I$$

$$V_1 V_1^T = ?$$

$$\Sigma^T = \begin{bmatrix} \Sigma_1^{-1} & 0 \\ 0 & 0 \end{bmatrix}$$

$$x = V_1 \alpha$$

$$= V_1 \Sigma_1^{-1} V_1^T b$$

$$= \underline{V \Sigma^+ U_1^T b}$$

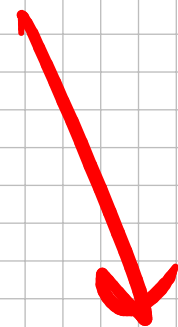
$$x = V_1 \alpha$$

$$= V_1 \Sigma_1^{-1} V_1^T b$$

$$= \underline{V \Sigma^+ U_1^T b}$$

$$A = U \Sigma V^T$$

$$A^+ = V \Sigma^+ U^T$$



$$x_{\text{SVD}} = \frac{1}{\sigma_1} v_1 (u_1^T b) + \frac{1}{\sigma_2} v_2 (u_2^T b) + \dots + \frac{1}{\sigma_r} v_r (u_r^T b)$$

$$x_{\text{SVD}} = x_{\text{LS}} = x^* = \bar{x} = \boxed{(A^T A)^{-1} A^T b}$$

• Notus SVD ✓

• Schor

• Valores Propios (Simetricas) →

SVD ←